Alexa VanHattum

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Research Interests

Programming languages + systems, compilers, formal methods

Teaching Interests

Intro. CS, Intro. Systems/OS, Programming Languages, Compilers, Software Engineering, Parallel Programming, Formal Methods Elective(s). With additional prep.: Distributed Systems, Networks, Databases

Education

Fall 2018–Present **Cornell University** | Ph.D. Candidate, Computer Science
Thesis: *Solver-aided Compilation for Systems Programming*
Expected graduation: May 2023

December 2021 **Cornell University** | M.S., Computer Science

2012–2016 **Brown University** | B.Sc., Computer Science

Publications

ICSE SEIP 2022 *Verifying Dynamic Trait Objects in Rust*
Alexa VanHattum, Daniel Schwartz-Narbonne, Nathan Chong, and Adrian Sampson. International Conference on Software Engineering, Software Engineering in Practice track (ICSE SEIP 2022). PDF. DOI.

ASPLOS 2021 *Vectorization for Digital Signal Processors via Equality Saturation.*
Alexa VanHattum, Rachit Nigam, Vincent T. Lee, James Bornholt, Adrian Sampson. International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2021). PDF. DOI.

LCTES 2020 *A Synthesis-aided Compiler for DSP Architectures (WiP Paper).*
Alexa VanHattum*, Rachit Nigam*, Vincent T. Lee, James Bornholt, Adrian Sampson (*equal contribution). Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES 2020). PDF. DOI.

Works in Progress

A Rewrite-based Approach to Verified Instruction Lowering.
To be submitted to PLDI 2023 or OSDI 2023. **Alexa VanHattum**, Monica Pardeshi, Chris Fallin, Adrian Sampson, Fraser Brown.

Selected Awards & Grants

April 2020–Present **NSF Graduate Research Fellow 2020**

April 2020 **Finalist: Qualcomm Innovation Fellowship 2020** with Rachit Nigam

May 2019	**Outstanding Teaching Assistant** from Cornell CS, for Programming Languages & Logics
April 2019	**Travel Scholarship** CRA-W Grad Cohort for Women Workshop

Teaching Experience

Summer 2022	**Object-Oriented Programming and Data Structures**, Cornell University, Instructor of Record 70 students, second-semester introductory computer science, taught in Java. As instructor, I (1) added programming labs, (2) added polling questions for active learning, and (3) shifted to frequent, lower-stakes exams.
Fall 2020	**Programming Languages & Logics**, Cornell University, Graduate TA
Fall 2018	**Programming Languages & Logics**, Cornell University, Graduate TA
Spring 2016	**Logic for Systems**, Brown University, Head TA
Fall 2017	**Intro. Object-Oriented Programming**, Brown University, Head TA
Spring 2015	**Discrete Structures & Probability**, Brown University, TA
Fall 2014	**Intro. Object-Oriented Programming**, Brown University, TA
Spring 2014	**First Byte of Computer Science**, Brown University, TA

Research Students Supervised

Spring '22-present	Monica Pardeshi, Carnegie Mellon post-undergraduate/masters, *A Rewrite-based Approach to Verified Instruction Lowering* → MS student at CMU
Fall '21-Spring '22	Alaia Solko-Breslin, Cornell undergraduate senior/5th year masters, *Synthesizing Chained Rust Calls* → PhD student at U. of Pennsylvania
Spring '21-Spring '22	Jasper Liang, Cornell undergraduate junior/senior, *Diospyros-LLVM: evaluating E-graph optimizations* → PhD student at Northwestern
Spring '21-Spring '22	Jonathan Tran, Cornell undergraduate senior/masters, *Diospyros-LLVM: control flow in E-graph optimizations* → MS student at Cornell
Summer '20	Jacob M. Delgado-López, University of Puerto Rico undergraduate first year, Louis Stokes Alliance for Minority Participation REU, *Benchmarks for synthesizing digital signal processing kernels* → subsequent REUs at Harvard, CMU
Fall '19-Spring '20	Katy Voor, Cornell undergraduate senior, *Scaling evaluation of spacial many-core compilation* → Meta Hack language team

Other Research Positions

May–Dec 2021	**Amazon Web Services**, Automated Reasoning Applied Scientist Intern. Mentors: Daniel Schwartz-Narbonne, Nathan Chong
Spring 2016	**Brown University**, Independent Study in Programming Languages. Advisor: Shriram Krishnamurthi

Summer 2014	**The University of Toledo**, Physics NSF REU. Advisor: Jaques G. Amar

Prior Industry Positions

2016–2018	**Apple**, Health Software Engineer. Lead app/database engineer for Apple's FDA-regulated heart rhythm feature.
Summer 2015	**Microsoft**, Software Engineering Intern

Invited Talks

April 2022	**Verifying Dynamic Trait Objects in Rust** Brown University
April 2022	**Verifying Dynamic Trait Objects in Rust** University of Texas at Austin
December 2021	**Vectorization for DSPs via Equality Saturation** Wellesley College
April 2021	**Vectorization for DSPs via Equality Saturation** Brown University
April 2021	**Guest Lectures: Finite State Machines, Graph Theory** Contra Costa Community College
December 2020	**Vectorization for DSPs via Equality Saturation** University of California Santa Cruz

Service & Outreach

Aug 2019–Present	**Graduate Students for Gender Inclusion in Computing** Founding member, executive board (current Vice President)
Mar–June 2022	**EGRAPHS Workshop at PLDI 2022** Program Committee (invited)
Summer 2020	**SPLASH OOPSLA 2020** Artifact Evaluation Committee
Aug 2020–Dec 2021	**Programming Languages Discussion Group** Co-organizer
Jan 2019–Dec 2021	**Cornell CS Colloquium** Student host organizer
Aug-Dec 2020	**CS Teaching Committee (re: COVID-19)** Student representative
June 2019	**PLDI 2019** Student volunteer
Spring 2019	**Cornell CS PhD Visit Day** Co-organizer
Spring 2019	**Cornell Girls' Adventures in Math** Student volunteer
2016–2017	**Apple Women in Science and Engineering Mentoring** Mentor